

Refractory epilepsy in a Chinese population

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Abstract

Objectives: To investigate the proportion of Chinese patients with intractable seizures and the risk factors leading to refractory epilepsy.

Methods: Consecutive patients over 14 years of age attending a Neurology clinic were evaluated. Patients with epilepsy were classified into two groups according to their seizure control: refractory or seizure-free. Epilepsy was classified as idiopathic as defined by age-related onset and typical electroclinical characteristics, symptomatic if secondary to a structural abnormality and cryptogenic if the cause was unknown. Age, sex, epilepsy syndrome classification, aetiology, presence of mental retardation and the number of drugs used were compared between patients with refractory epilepsy and those in remission.

Results: Among 260 adolescent and adult patients with a mean age of 34 years (range 15–79), complete seizure control was achieved in 157 (60%) cases. Multivariate binomial logistic regression analysis showed that patients with mesial temporal sclerosis (OR = 7.6, 95% CI 3.53–16.4, $p < 0.01$) and the presence of mental retardation (OR = 9.39, 95% CI 3.98–22.12, $p < 0.01$) were more likely to develop pharmacoresistant epilepsy.

Conclusion: In adults the underlying aetiology is an important factor as to whether patients develop intractable seizures. Poor control was also associated with the presence of mesial temporal sclerosis and mental retardation.

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Keywords: Refractory epilepsy; Magnetic resonance imaging; Mesial temporal sclerosis; Seizures

1. Introduction

Epilepsy is a heterogeneous condition with different causes and outcomes. The majority of epileptic individuals live in Asia but there is very little data on the prognoses of patients from these countries [1]. Extrapolating from conservative estimates, there are over six million sufferers in China alone [2–4]. Hong Kong is in a good position to study the characteristics of this problem as 97% of the residents are ethnic Chinese and there is a comprehensive health care system that allows detailed evaluation of patients. We studied a cohort of patients with epilepsy in order to examine the proportion of patients with intractable seizures and the risk factors leading to refractory epilepsy in a Chinese population.

2. Methods

2.1. Patients

We prospectively evaluated unselected patients over the age of 14 years with epilepsy from a Neurology clinic at a regional hospital, the Prince of Wales hospital in Hong Kong, China that caters for a population of 800,000. Patient data had been collected from a registry since 1997 [5]. Cases were referred to this institution from a number of different sources: primary care physicians, other hospital departments and the accident and emergency unit. Patients in our clinic included individuals already on anti-epileptic drugs (AED) as well those who were newly diagnosed with epilepsy. Locally most people with chronic neurological illnesses are cared for by the government's Hospital Authority system of which this hospital is part. Relevant information was entered

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into a database by a single neurologist: age, sex, clinical history and physical examination results, seizure type and epilepsy syndrome, duration of follow-up, the presence of mental retardation, aetiology of epilepsy, anti-epileptic drug (AED) usage, results of magnetic resonance imaging (MRI), electroencephalography and adequacy of seizure control. Cases with single unprovoked seizures and situation-related seizures were excluded, as were those with a diagnosis other than epilepsy such as psychogenic seizures.

2.2. Definitions

Seizure types were categorized, according to the International League Against Epilepsy (ILAE) classification of epileptic seizures, into partial (simple or complex partial) or generalized (e.g., absence, myoclonic, tonic, atonic, clonic, tonic-clonic) [6]. Epilepsy syndrome was classified as idiopathic, symptomatic, or cryptogenic, using electroclinical criteria as defined by the ILAE classification of epilepsies and epileptic syndromes [7]. Idiopathic epilepsies (such as juvenile myoclonic epilepsy) were presumed to be genetic in origin and the diagnosis was based on typical clinical and EEG data. Cryptogenic epilepsy was presumed to be the consequence of an underlying but undetermined abnormality. Symptomatic epilepsy was considered to be secondary to a known structural lesion such as cortical dysplasia or tumor. A diagnosis of symptomatic epilepsy was made when there was radiological or clinical evidence of brain damage secondary to the ascribed insult such as cerebral infarct or meningitis. For example in someone with the clinical and EEG evidence of temporal lobe epilepsy, if the MRI had demonstrated hippocampal atrophy the aetiology would be regarded as mesial temporal sclerosis. As a part of standard care at our clinic, basic laboratory investigation such as blood glucose, renal and liver function tests, surface electroencephalogram as well as neuroimaging were performed in patients. MRI of the brain was performed using a high resolution 1.5 T Philips Gyroscan ACS-NT. Standardised MR protocols consisted of (a) T₁-weighted spin-echo axial images (repetition time (TR)=500 ms; echo time (TE)=15 ms; slice thickness=5 mm; 0.5 mm gap; flip angle=90; field of view (FOV)=230 mm); (b) T₂-weighted turbo-spin-echo coronal images (TR=3300 ms; TE=100 ms; slice thickness=5 mm; no gap; flip angle=90; FOV=230 mm) and (c) T₂-weighted turbo-spin-echo oblique coronal images with image plane perpendicular to the longitudinal axis of the hippocampus (TR=3300 ms; TE=100 ms; slice thickness=2 mm; no gap; flip angle=90; FOV=230 mm). MR films were interpreted and reviewed by a neurologist and a radiologist. Lesions shown on MR imaging were classified as neuronal migration disorders (e.g., grey matter heterotopia, tuberous sclerosis, focal cortical dysplasia), mesial temporal sclerosis (MTS); neocortical sclerosis (evidence of prior CNS insult such as CNS infection, trauma, cerebrovascular disease), tumors, vascular malformations (cavernous angioma, arteriovenous malformations) and other abnormalities. The majority of

cases with electroclinical evidence for idiopathic epilepsies did not have MRI performed routinely as it is generally accepted that this would not be part of the routine work-up of these individuals. Only a small proportion of cases underwent more intensive investigations such as MR spectroscopy, functional MRI and SPECT and so information from these findings were not used for the purposes of this study.

In our clinic at each visit, seizure frequency, response to drug therapy and compliance were routinely recorded. Anti-epileptic treatment consisted of the most appropriate drug after considering the seizure and syndromic classification, patient profile and drug characteristics. The number of AED that had been prescribed sequentially in order to control seizures was recorded. Local first line drugs included carbamazepine, valproate, phenytoin and phenobarbitone while of the newer drugs, vigabatrin, lamotrigine, gabapentin and topiramate are available. AED are continued for a minimum of 3 years, after which the drugs are tapered off if the patient is suitable for drug withdrawal, following the guidelines recommended by the American Academy of Neurology [8]. During the drug free period, they would be then followed for 1 year before discharge. Refractory epilepsy was defined as the presence of one or more seizures despite the use of two appropriate AEDs. For patients who went on to have epilepsy surgery or vagus nerve stimulation the seizure frequency prior to these procedures were used to determine epilepsy control.

2.3. Statistics

SPSS 11 software for Windows (Chicago, IL) was used to carry out all of the analyses.

Demographic data including age, sex, epilepsy classification, aetiology, presence of mental retardation and the number of drugs that had been tried were compared between patients with refractory epilepsy and those in remission. Categorical data were analyzed using the χ^2 -test; Fisher's Exact Method was used when the expected count of any cell in the 2×2 table was less than five. Continuous data were compared using independent *t*-tests. The above variables were analyzed separately with univariate binomial logistic regression analyses; patients were categorized into either a refractory or seizure-free group and this was entered as the dependent variable. Significant variables in the univariate analyses were then submitted to a multivariate binomial logistic model to examine their independent contribution to the dependent variable. Statistical significance was set at $p < 0.05$.

3. Results

The series comprised 260 patients, 127 (49%) men and 133 (51%) women with a mean age of 34 years (range 15–79 years). Out of these, 161 (63%) were newly diagnosed and 99 (37%) had already started treatment with AED at another clinic or hospital. The mean duration of follow-up was 7

Table 1

Comparison of characteristics between seizure-free and refractory patients

	Seizure-free	Refractory
N	157	103
Age	35.4 (S.D. 11.8)	33.9 (S.D. 9.5)
Gender (% female)	54	47
Aetiology		
Unknown	53 (34%)	22 (21%)
Dysplasia	5 (3%)	8 (8%)
Tumor	4 (3%)	5 (5%)
Vascular	6 (4%)	7 (7%)
Stroke	16 (10%)	6 (6%)
Idiopathic	23 (15%)	3 (3%)
MTS	8 (5%)	28 (27%)
CNS infection	28 (17%)	16 (16%)
Scar	4 (3%)	4 (4%)
RTA	10 (6%)	4 (4%)
ILAE		
Idiopathic	23 (15%)	3 (3%)
Symptomatic	85 (54%)	80 (78%)
Cryptogenic	49 (31%)	20 (19%)
Mental retardation	11 (7%)	31 (30%)
No. of drugs	1.2 (0.6)	1.8 (0.8)

MTS: mesial temporal sclerosis; CNS: central nervous system; RTA: road traffic accident; ILAE: International League Against Epilepsy.

years (range 1–17 years). At the time of analysis 157 (60%) patients were in remission and the characteristics of cases in both refractory and seizure-free groups are shown in Table 1. All patients with cryptogenic and localization-related epilepsies had cranial MRI performed except for those with severe mental retardation, claustrophobia or contraindication for MR (such as presence of non-compatible pacemaker); therefore cranial computed tomography (CT) were done for these cases. There were five such patients and their CT scans support clinical evidence of previous cerebral infarction ($n=2$), CNS infection ($n=2$) and post-traumatic epilepsy ($n=1$). MRI in 75 (29%) patients were normal. Patients with neuronal migration disorders were not further subdivided because of the small numbers involved. Univariate analysis showed that patients who were prescribed more drugs and who had symptomatic epilepsy were more likely to continue to have seizures while those with idiopathic epilepsy tended to have better control (Table 2). Multivariate binomial logistic regression analysis showed that among the factors analysed, patients with mesial temporal sclerosis (OR = 7.6, 95% CI 3.53–16.4, $p<0.01$) and the presence of mental retardation (OR = 9.39, 95% CI 3.98–22.12, $p<0.01$) were associated with refractory epilepsy.

4. Discussion

Based predominantly on studies of patients with epilepsy from the UK, Europe and the US, a number of risk factors for refractory epilepsy have been identified: seizure type, the number of seizures at the early phase of disease, psychiatric deficits, electro-encephalographic abnormalities, structural

Table 2

Univariate regression analysis for predictors for refractory epilepsy

	95% CI			
	OR	Lower	Upper	<i>p</i>
Age	0.99	0.96	1.01	0.29
Gender	1.35	0.82	2.22	0.23
Cause				
Unknown	0.53	0.3	0.95	0.32
Dysplasia	2.56	0.81	8.05	0.10
Tumor	1.95	0.51	7.44	0.33
Vascular	1.83	0.59	5.62	0.29
Stroke	0.54	0.2	1.44	0.22
Idiopathic	0.17	0.05	0.59	<0.01
MTS	6.95	3.02	15.99	<0.01
CNS infection	0.84	0.43	1.65	0.63
Scar	1.54	0.37	6.32	0.55
RTA	0.59	0.18	1.94	0.39
ILAE				
Idiopathic	0.17	0.05	0.59	<0.01
Symptomatic	2.94	1.68	5.16	<0.01
Cryptogenic	0.53	0.29	0.96	0.04
Mental retardation	5.71	2.71	12.02	<0.01
No. of drugs	3.66	2.38	5.62	<0.01

MTS: mesial temporal sclerosis; CNS: central nervous system; RTA: road traffic accident; ILAE: International League Against Epilepsy.

lesions on imaging and inadequate response to initial drug treatment [9–26]. Whether these apply to other ethnic groups is uncertain; this is the first report that attempts to identify predictors associated with intractable seizures in a Chinese population. We found that the proportion of individuals who were seizure-free were similar to estimates from recent hospital based studies, between 43 and 63% [21,24,26]. In this series of those who were seizure free, 78% were on a single drug; one earlier report had found that 47% were in remission on monotherapy and 90% of these required moderate doses only [24]. The results from this study support the view that for most patients, good control with a single agent is possible.

Four points should be noted. First, data from hospital-based centres are subject to referral bias, as more cases with refractory epilepsy are expected than from primary care physicians. Second, our patients excluded children aged 14 and below. The aetiology, outcome and prognosis in paediatric patients would be expected to differ from adults. Third, the percentage of people in remission also depends on the definition used; we used a previously published definition of refractory epilepsy as the criterion [21,24,26]. Fourth, we were unable in all cases to determine accurately the number of seizures before treatment was started and therefore did not incorporate this variable into the analysis.

Overall the aetiological profile in this group of patients was similar to other hospital-based populations but epilepsy due to previous central nervous system infections was more common in our region. Although patients with idiopathic epilepsy tended to have better control this was not the case after multivariate analysis but this should be interpreted with caution due to the small number of cases with this diagnosis.

In keeping with previous studies, we found that the majority of patients with MTS (78%) had intractable seizures [20,21]. However, a small proportion of patients with MTS remained in remission, suggesting that some cases have a more benign form. Among the information that could be obtained early during the evaluation of patients we found that the presence of mental retardation and multiple seizure types were poor prognostic factors [14,25].

In conclusion the outcome for patients with epilepsy is variable; recognizing those who are at risk of developing intractable seizures an early stage would help in the management of refractory epilepsy.

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